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Effects of external electric field and anisotropic long-range reactivity on charge separation probability

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We consider the effects of external electric field and anisotropic long-range reactivity on the recombination dynamics of a geminate charge pair. A closed-form analytic expression for the ultimate separation probability of the pair is presented. In previous theories, analytic expressions for the separation probability were obtained only for the case where the recombination reaction can be assumed to occur at a contact separation. For this case, Noolandi and Hong obtained an exact solution, but their expression for the separation probability was too complicated to evaluate. Hence an approximate analytic expression proposed by Braun has been widely used. However, Braun's expression overestimates the separation probability when the electric field is large. In this work, we present an approximate analytic expression that is accurate enough for all parameter values. In addition, the expression is also applicable when the interaction between the geminate charge pair is described by screened Coulombic potential, and the recombination reaction has an anisotropic and long-range reactivity. We also provide the expression for the separation probability when the initial separation between the geminate charge pair is larger than the contact distance. *Published by AIP Publishing.*
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I. INTRODUCTION

The recombination of geminate charge pairs, produced by radiation or electron bombardment, was first considered by Onsager.[1] He derived an expression for the ultimate separation probability of a geminate pair, undergoing the Brownian motion in the presence of an external electric field as well as the attractive Coulomb interaction between the charge pair. Noolandi and Hong[2] extended the Onsager theory to the case of the finite recombination rate at nonzero reaction radius. Although they obtained an exact expression for the separation probability of a geminate pair, it involved an infinite series of product terms of two complicated functions, each of which was expressed also as an infinite series. Furthermore, the involved expansion coefficients need to be determined by solving a set of linear equations containing infinite sums. As such, a semi-empirical expression for the charge separation probability proposed by Braun[3] has been more widely used instead.

However, as shown by Wojcik and Tachiya,[4] Braun's expression severely overestimates the external electric field effect on the separation probability. Wojcik and Tachiya devised a simple correction to Braun's expression, which provides very accurate results unless either the external electric field or the inherent recombination rate is large. Very recently, Seki and Wojcik[5] obtained an approximate expression for the charge separation probability that is much more accurate than the corrected Braun's expression over the extended range of external electric field strength. However, the validity of their expression is limited to the case with large Onsager distance so that the expression is useful only when the reaction medium has low electrical permittivity.

The previous expressions for the charge separation probability were obtained only for the case where the recombination reaction can be assumed to occur at a contact separation. In this work, we employ the recently proposed solution method[6,7] for Fredholm integral equations of the second kind to treat the effects of external electric field and anisotropic long-range reactivity on the recombination dynamics of a geminate charge pair. A closed-form analytic expression for the ultimate separation probability of the pair is presented, which is accurate enough for all parameter values. As in Refs. 2 and 5, we also consider the case when the initial separation between the geminate charge pair is larger than the contact distance. In addition, the case in which the interaction between the charge pair is given by arbitrary central potential is also considered.

II. THEORY

A. Reaction-diffusion model

We consider the recombination of a geminate pair of charged particles that are generated initially at a relative position $\mathbf{r}_0$. We assume that thermal motions of the particles may be described by the Smoluchowski equation, and that the particles at a relative position $\mathbf{r}$ recombine at a rate $S_R(\mathbf{r})$. Then the probability density $G_R(\mathbf{r}, t|\,\mathbf{r}_0)$ of finding the pair of particles at a relative position $\mathbf{r}$ at time t evolves in time as

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$$\frac{\partial}{\partial t}G_R(\mathbf{r},t\,|\mathbf{r}_0) = \left(\frac{\partial}{\partial \mathbf{r}}\right)^T \cdot De^{-U(\mathbf{r})}\frac{\partial}{\partial \mathbf{r}}e^{U(\mathbf{r})}G_R(\mathbf{r},t\,|\mathbf{r}_0) - S_R(\mathbf{r})G_R(\mathbf{r},t\,|\mathbf{r}_0). \quad (1)$$

We neglect the hydrodynamic interaction between the geminate particles so that the relative diffusion coefficient D is assumed to be given by the sum of their translational diffusion constants. $U(\mathbf{r})$ denotes the potential energy divided by k_BT (k_B and T are Boltzmann's constant and the absolute temperature). $\partial/\partial\mathbf{r}$ is the del operator, denoting a column vector of three partial derivative operators, and $(\partial/\partial\mathbf{r})^T$ denotes its transpose.

The potential of mean force between the geminate particles in the absence of the external electric field is denoted by $U_1(r)$. We assume that $U(\mathbf{r})$ is simply given by the sum of $U_1(r)$ and the potential energy due to the external electric field. Denoting the relative position of the positively charged particle with respect to the negatively charged one by $\mathbf{r}$ and choosing the z-axis to lie in the direction of the external electric field $\mathbf{E}$, we can then write

$$U(\mathbf{r}) = U(r,\mu) = U_1(r) - Kr\mu, \quad (2)$$

where $K = eE/k_BT$ and $\mu = \cos\theta$ with $E = |\mathbf{E}|$ and $\cos\theta = \mathbf{r}\cdot\mathbf{E}/|\mathbf{r}|\,|\mathbf{E}|$. e denotes the magnitude of the charges of the geminate particles produced by radiation, which should usually equal to the proton charge.

The quantity of interest is the survival probability of the geminate particles that are still free by time t. The survival probability $W(\mathbf{r}_0,t)$ of the geminate particles, whose relative position is $\mathbf{r}_0$ at $t = 0$, is related to $G_R(\mathbf{r}, t|\,\mathbf{r}_0)$ as $W(\mathbf{r}_0,t) = \int d\mathbf{r}\, G_R(\mathbf{r},t\,|\mathbf{r}_0)$. Hence, its evolution equation follows directly from the adjoint equation of the Smoluchowski equation, which can be in turn derived from Eq. (1) by using the detailed balance condition,[8]

$$G_R(\mathbf{r},t\,|\mathbf{r}_0)e^{-U(\mathbf{r}_0)} = G_R(\mathbf{r}_0,t\,|\mathbf{r})e^{-U(\mathbf{r})}. \quad (3)$$

We have

$$\frac{\partial}{\partial t}W(\mathbf{r}_0,t) = e^{U(\mathbf{r}_0)}\left(\frac{\partial}{\partial \mathbf{r}_0}\right)^T \cdot De^{-U(\mathbf{r}_0)}\frac{\partial}{\partial \mathbf{r}_0}W(\mathbf{r}_0,t) - S_R(\mathbf{r}_0)W(\mathbf{r}_0,t). \quad (4)$$

Owing to the symmetry of the problem, $W(\mathbf{r}_0,t)$ is independent of the azimuth angle. Further, if we denote the radial and polar angles of $\mathbf{r}_0$ simply by (r,θ) rather than by (r_0,θ_0), the evolution equation for $W(\mathbf{r}_0,t) = W(r,\mu = \cos\theta,t)$ takes the form

$$\frac{\partial}{\partial t}W(r,\mu,t) = \frac{De^{U(r,\mu)}}{r^2}\frac{\partial}{\partial r}\left[r^2e^{-U(r,\mu)}\frac{\partial}{\partial r}W(r,\mu,t)\right] + \frac{De^{U(r,\mu)}}{r^2}\frac{\partial}{\partial \mu}\left[(1-\mu^2)\,e^{-U(r,\mu)}\frac{\partial}{\partial \mu}W(r,\mu,t)\right] - S_R(r,\mu)W(r,\mu,t). \quad (5)$$

The solution to the full time-dependent problem is very complicated.[9] In this work, we restrict ourselves to obtaining an approximate expression for the survival probability in the steady state attained at long times. This *ultimate* survival probability $W_u(r,\mu)$, which may also be called the charge separation probability, satisfies the following equation:

$$0 = \frac{\partial}{\partial r}\left[r^2e^{-U(r,\mu)}\frac{\partial}{\partial r}W_u(r,\mu)\right] + \frac{\partial}{\partial \mu}\left[(1-\mu^2)\,e^{-U(r,\mu)}\frac{\partial}{\partial \mu}W_u(r,\mu)\right] - \frac{r^2S_R(r,\mu)}{De^{U(r,\mu)}}W_u(r,\mu). \quad (6)$$

The boundary conditions associated with the present problem are

$$\lim_{r\to\infty} W_u(r,\mu) = 1, \quad (7)$$

$$\left.\frac{\partial}{\partial r}W_u(r,\mu)\right|_{r=\sigma} = 0, \quad (8)$$

where σ denotes the contact distance between the geminate particles. The outer boundary condition in Eq. (7) must be physically self-evident. The inner boundary condition in Eq. (8) follows from the condition that the radial probability flux is zero at σ,

$$J_r(r=\sigma,\mu,t\,|r_0\,,\mu_0) = -De^{-U(\sigma,\mu)}\frac{\partial}{\partial r}e^{U(r,\mu)}G_R(r,\mu,t\,|r_0\,,\mu_0)\Big|_{r=\sigma} = 0. \quad (9)$$

B. Solution to Eq. (6) for a δ-function reaction sink

For simplicity, we will assume that the initial distribution of the geminate charge pair is isotropic so that the quantity we have to evaluate is the orientation-averaged survival probability,

$$\bar{W}_u(r) = \frac{1}{2}\int_{-1}^{1} d\mu\; W_u(r,\mu). \quad (10)$$

Then, integrating Eq. (6) over the μ coordinate, we obtain

$$\frac{\partial}{\partial r}\left[r^2\frac{1}{2}\int_{-1}^{1} d\mu\, e^{-U(r,\mu)}\frac{\partial}{\partial r}W_u(r,\mu)\right] = \frac{r^2}{D}\frac{1}{2}\int_{-1}^{1} d\mu\frac{S_R(r,\mu)}{e^{U(r,\mu)}}W_u(r,\mu). \quad (11)$$

Next, by integrating Eq. (11) over r from σ to r_1 and using the inner boundary condition in Eq. (8), we get

$${r_1}^2\frac{1}{2}\int_{-1}^{1} d\mu\, e^{-U(r_1,\mu)}\frac{\partial}{\partial r_1}W_u(r_1,\mu) = \int_{\sigma}^{r_1} dr\frac{r^2}{D}\frac{1}{2}\int_{-1}^{1} d\mu\frac{S_R(r,\mu)}{e^{U(r,\mu)}}W_u(r,\mu). \quad (12)$$

Let us then consider the simpler case in which the recombination reaction can occur only at the contact distance σ. In this case, the reaction sink function can be expressed in the form

$$S_R(r,\mu) = \kappa(\mu)\delta(r-\sigma)/(4\pi\sigma^2), \quad (13)$$

where $\kappa(\mu)$ denotes an orientation-dependent inherent rate coefficient. For the sink function in Eq. (13) and the potential energy given by Eq. (2), Eq. (12) can be rewritten as

$$\frac{1}{2}\int_{-1}^{1} d\mu\, e^{Kr_1\mu}\frac{\partial}{\partial r_1}W_u(r_1,\mu) = \left[\frac{e^{-U_1(\sigma)}}{4\pi D}\frac{1}{2}\int_{-1}^{1} d\mu\, e^{K\sigma\mu}\kappa(\mu)W_u(\sigma,\mu)\right]\frac{e^{U_1(r_1)}}{{r_1}^2}. \quad (14)$$

Since $e^{Kr_1\mu}$ and $e^{K\sigma\mu}\kappa(\mu)$ are positive for all values of μ, by applying the mean value theorem, we have

$$\begin{aligned}\frac{1}{2}\int_{-1}^{1} d\mu\, e^{Kr_1\mu}\frac{\partial}{\partial r_1}W_u(r_1,\mu)\\ &= \left[\frac{1}{2}\int_{-1}^{1} d\mu\, e^{Kr_1\mu}\right]\frac{\partial}{\partial r_1}W_u(r_1,\mu_1^*)\\ &= \frac{\sinh Kr_1}{Kr_1}\frac{\partial}{\partial r_1}W_u(r_1,\mu_1^*),\end{aligned} \tag{15}$$

$$\begin{aligned}\frac{1}{2}\int_{-1}^{1} d\mu\, e^{K\sigma\mu}\kappa(\mu)W_u(\sigma,\mu)\\ &= \left[\frac{1}{2}\int_{-1}^{1} d\mu\, e^{K\sigma\mu}\kappa(\mu)\right]W_u(\sigma,\mu_0^*),\end{aligned} \tag{16}$$

where μ_0^* and μ_1^* are some values of μ in the interval $(-1, 1)$. Then, Eq. (14) becomes

$$\frac{\sinh Kr_1}{Kr_1}\frac{\partial}{\partial r_1}W_u(r_1,\mu_1^*) = \left[\frac{e^{-U_1(\sigma)}}{4\pi D}\Lambda_{rx}W_u(\sigma,\mu_0^*)\right]\frac{e^{U_1(r_1)}}{{r_1}^2}, \tag{17}$$

where

$$\Lambda_{rx} \equiv \frac{1}{2}\int_{-1}^{1} d\mu\, e^{K\sigma\mu}\kappa(\mu). \tag{18}$$

We will now make some critical approximations. If $W_u(r,\mu)$ does not depend much on μ, we may make the following approximations:

$$\frac{\partial}{\partial r}W_u(r,\mu_1^*) \cong \frac{\partial}{\partial r}\bar{W}_u(r), \tag{19}$$

$$W_u(\sigma,\mu_0^*) \cong \bar{W}_u(\sigma), \tag{20}$$

where $\bar{W}_u(r)$ is the orientation-averaged survival probability defined by Eq. (10). The conditions for the validity of these approximations will be described in Sec. III in physical terms. With Eqs. (19) and (20), Eq. (17) can be rewritten as

$$\frac{\partial}{\partial r_1}\bar{W}_u(r_1) = \left[\frac{e^{-U_1(\sigma)}}{4\pi D}\Lambda_{rx}\bar{W}_u(\sigma)\right]\frac{Kr_1}{\sinh Kr_1}\frac{e^{U_1(r_1)}}{{r_1}^2}. \tag{21}$$

Finally, by integrating Eq. (21) over r_1 from r to infinity and using the outer boundary condition in Eq. (7), we get

$$1-\bar{W}_u(r) = \left[\frac{e^{-U_1(\sigma)}}{4\pi D}\Lambda_{rx}\right]\left[\int_r^{\infty} dr_1\frac{Kr_1}{\sinh Kr_1}\frac{e^{U_1(r_1)}}{{r_1}^2}\right]\bar{W}_u(\sigma). \tag{22}$$

The integral appearing on the right hand side of Eq. (22) is a kind of special function that will appear repeatedly throughout this paper. We will denote the integral in terms of a dimensionless function $\chi(z)$ as

$$\int_r^{\infty} dr_1\frac{Kr_1}{\sinh Kr_1}\frac{e^{U_1(r_1)}}{{r_1}^2} = \frac{1}{\sigma}\chi(\sigma/r),$$

$$\chi(z) \equiv \int_0^{z} dy\frac{K\sigma/y}{\sinh(K\sigma/y)}e^{U_1(\sigma/y)}. \tag{23}$$

By setting $r = \sigma$ in Eq. (22), we have

$$\bar{W}_u(\sigma) = \left[1+\frac{e^{-U_1(\sigma)}}{4\pi D\sigma}\Lambda_{rx}\chi(1)\right]^{-1}, \tag{24}$$

where Λ_{rx} and $\chi(z)$ are defined by Eqs. (18) and (23), respectively. With this expression for $\bar{W}_u(\sigma)$, Eq. (22) gives

$$\bar{W}_u(r) = \left\{1+\frac{e^{-U_1(\sigma)}}{4\pi D\sigma}\Lambda_{rx}[\chi(1)-\chi(\sigma/r)]\right\}\bar{W}_u(\sigma). \tag{25}$$

C. Solution to Eq. (6) for long-range reaction sink functions

We now consider the more general case in which the recombination reaction can occur at a range of separations between geminate reactant particles. It is noted that Eq. (12) still holds. Since $e^{-U(r_1,\mu)}$ and $S_R(r,\mu)/e^{U(r,\mu)}$ are positive for all values of μ, we apply the mean value theorem to the integrals appearing in Eq. (12) to get

$$\begin{aligned}{r_1}^2\left[\frac{1}{2}\int_{-1}^{1} d\mu\, e^{-U(r_1,\mu)}\right]\frac{\partial}{\partial r_1}W_u(r_1,\mu_1^*)\\ &= \int_{\sigma}^{r_1} dr\frac{r^2}{D}\left[\frac{1}{2}\int_{-1}^{1} d\mu\frac{S_R(r,\mu)}{e^{U(r,\mu)}}\right]W_u(r,\mu_0^*),\end{aligned} \tag{26}$$

where μ_0^* and μ_1^* are some values of μ in the interval $(-1, 1)$. For the potential energy given by Eq. (2), Eq. (26) can be rewritten as

$$\begin{aligned}{r_1}^2 e^{-U_1(r_1)}\frac{\sinh Kr_1}{Kr_1}\frac{\partial}{\partial r_1}W_u(r_1,\mu_1^*)\\ &= \frac{1}{D}\int_{\sigma}^{r_1} dr\frac{r^2\Lambda(r)}{e^{U_1(r)}}W_u(r,\mu_0^*),\end{aligned} \tag{27}$$

where $\Lambda(r)$ is defined by

$$\Lambda(r) \equiv \frac{1}{2}\int_{-1}^{1} d\mu\, e^{Kr\mu}S_R(r,\mu). \tag{28}$$

We then make the critical approximations as given by Eqs. (19) and (20) to rewrite Eq. (27) as

$$\frac{\partial}{\partial r_1}\bar{W}_u(r_1) = \frac{1}{D}\frac{Kr_1}{\sinh Kr_1}\frac{e^{U_1(r_1)}}{{r_1}^2}\int_{\sigma}^{r_1} dr_2\frac{r_2^2\Lambda(r_2)}{e^{U_1(r_2)}}\bar{W}_u(r_2). \tag{29}$$

Next, by integrating Eq. (29) over r_1 from r to infinity and using the outer boundary condition in Eq. (7), we get

$$\begin{aligned}1-\bar{W}_u(r) &= \frac{1}{D}\int_r^{\infty} dr_1\frac{Kr_1}{\sinh Kr_1}\frac{e^{U_1(r_1)}}{{r_1}^2}\\ &\quad\times\int_{\sigma}^{r_1} dr_2\frac{r_2^2\Lambda(r_2)}{e^{U_1(r_2)}}\bar{W}_u(r_2).\end{aligned} \tag{30}$$

Integration by parts gives

$$\begin{aligned}\int_r^{\infty} dr_1\frac{Kr_1}{\sinh Kr_1}\frac{e^{U_1(r_1)}}{{r_1}^2}\int_{\sigma}^{r_1} dr_2\frac{r_2^2\Lambda(r_2)}{e^{U_1(r_2)}}\bar{W}_u(r_2)\\ &= \frac{1}{\sigma}\chi(\sigma/r)\int_{\sigma}^{r} dr_1\frac{r_1^2\Lambda(r_1)}{e^{U_1(r_1)}}\bar{W}_u(r_1)\\ &\quad+\int_r^{\infty} dr_1\frac{1}{\sigma}\chi(\sigma/r_1)\frac{r_1^2\Lambda(r_1)}{e^{U_1(r_1)}}\bar{W}_u(r_1),\end{aligned} \tag{31}$$

where $\chi(z)$ is the function defined in Eq. (23).

From Eqs. (30) and (31), we obtain the following integral equation for $\bar{W}_u(r)$:

$$\bar{W}_u(r) = 1 - \frac{\chi(\sigma/r)}{D\sigma}\int_\sigma^r dr_1 \frac{r_1^2\Lambda(r_1)}{e^{U_1(r_1)}}\bar{W}_u(r_1) - \frac{1}{D\sigma}\int_r^\infty dr_1\, \chi(\sigma/r_1)\frac{r_1^2\Lambda(r_1)}{e^{U_1(r_1)}}\bar{W}_u(r_1). \quad (32)$$

This type of integral equations is called the Fredholm integral equation of the second kind. An efficient solution method for this type of integral was proposed by us in Refs. 6 and 7. First, a *formally exact* solution to Eq. (32) is obtained as

$$\bar{W}_u(r) = \left[1 + \frac{\chi(\sigma/r)}{D\sigma}\int_\sigma^r dr_1 \frac{r_1^2\Lambda(r_1)}{e^{U_1(r_1)}}\frac{\bar{W}_u(r_1)}{\bar{W}_u(r)} + \frac{1}{D\sigma}\int_r^\infty dr_1\, \chi(\sigma/r_1)\frac{r_1^2\Lambda(r_1)}{e^{U_1(r_1)}}\frac{\bar{W}_u(r_1)}{\bar{W}_u(r)}\right]^{-1}. \quad (33)$$

Then the unknown ratio $\bar{W}_u(r_1)/\bar{W}_u(r)$ may be approximated by the truncated series solution of Eq. (32) or by some physically reasonable estimates. The former approach is more systematic in that a higher-order solution of Eq. (32) will provide a better estimate for the ratio $\bar{W}_u(r_1)/\bar{W}_u(r)$ in Eq. (33), but it would lead to a more involved solution. Because an expression for the ultimate survival probability is already available for the case with δ-function sinks [Eq. (25)], we thus make the following approximation:

$$\bar{W}_u(r_1)/\bar{W}_u(r) \cong \bar{W}_u^\delta(r_1)/\bar{W}_u^\delta(r). \quad (34)$$

Here, $\bar{W}_u^\delta(r)$ is the ultimate survival probability for a system with a delta-function reaction sink with the same contracted reactivity as the sink function $S_R(r,\mu)$ under consideration, that is,

$$\int d\mathbf{r}\, \frac{\kappa(\mu)\delta(r-\sigma)}{4\pi\sigma^2}e^{-U_1(r)} = \int d\mathbf{r}\, S_R(r,\mu)e^{-U_1(r)}. \quad (35)$$

Solving this equation for $\kappa(\mu)$, we get

$$\kappa(\mu) = 4\pi e^{U_1(\sigma)}\int_0^\infty dr\, r^2 S_R(r,\mu)e^{-U_1(r)}. \quad (36)$$

Using the expression for $\bar{W}_u^\delta(r)$ in Eq. (25), we approximate the ratio $\bar{W}_u(r_1)/\bar{W}_u(r)$ as

$$\frac{\bar{W}_u(r_1)}{\bar{W}_u(r)} \cong \frac{\bar{W}_u^\delta(r_1)}{\bar{W}_u^\delta(r)} = \frac{1 + \frac{e^{-U_1(\sigma)}}{4\pi D\sigma}\Lambda_{rx}[\chi(1)-\chi(\sigma/r_1)]}{1 + \frac{e^{-U_1(\sigma)}}{4\pi D\sigma}\Lambda_{rx}[\chi(1)-\chi(\sigma/r)]}, \quad (37)$$

where Λ_{rx} is given by

$$\Lambda_{rx} \equiv 4\pi e^{U_1(\sigma)}\int_0^\infty dr\, r^2 e^{-U_1(r)}\frac{1}{2}\int_{-1}^1 d\mu\, e^{K\sigma\mu}S_R(r,\mu). \quad (38)$$

Obviously, the accuracy of the approximation given by Eq. (34) gets worse when the reaction zone becomes very broad. However, the sink function for electron transfer reactions, which is of interest in this work, decays exponentially with r and is short-ranged in most cases.

To summarize, the ultimate survival probability is approximately given by

$$\bar{W}_u(r) = \left[1 + \frac{\chi(\sigma/r)}{D\sigma}\int_\sigma^r dr_1 \frac{r_1^2\Lambda(r_1)}{e^{U_1(r_1)}}\frac{\bar{W}_u^\delta(r_1)}{\bar{W}_u^\delta(r)} + \frac{1}{D\sigma}\int_r^\infty dr_1\, \chi(\sigma/r_1)\frac{r_1^2\Lambda(r_1)}{e^{U_1(r_1)}}\frac{\bar{W}_u^\delta(r_1)}{\bar{W}_u^\delta(r)}\right]^{-1}, \quad (39)$$

where the ratio $\bar{W}_u^\delta(r_1)/\bar{W}_u^\delta(r)$ is given by Eqs. (37) and (38), and the functions $\chi(z)$ and $\Lambda(r)$ are defined by Eqs. (23) and (28), respectively.

III. RESULTS AND DISCUSSION

A. Systems involving a δ-function reaction sink

In previous theories, analytic expressions for the ultimate survival probability were obtained only for the case when the geminate particles interact via Coulombic potential and the recombination reaction occurs at a contact separation with orientation-independent reactivity. This corresponds to the case in which $S_R(r,\mu)$ is given by the δ-function reaction sink in Eq. (13) with constant κ independent of μ and $U_1(r)$ is given by

$$U_1(r) = -r_c/r. \quad (40)$$

Here, the Onsager distance r_c is given by

$$r_c = e^2/(4\pi\varepsilon_0\varepsilon_r k_B T), \quad (41)$$

where ε_0 and ε_r denote the vacuum permittivity and the relative permittivity of the reaction medium, respectively.

In this case, Λ_{rx} defined in Eq. (18) is given by $\kappa(\sinh K\sigma)/(K\sigma)$, and from Eqs. (24) and (25) the separation probability of a geminate charge pair with initial separation r is given by

$$\bar{W}_u(\sigma) = \left[1 + \frac{\kappa e^{r_c/\sigma}}{4\pi D\sigma}\frac{\sinh K\sigma}{K\sigma}\chi(1)\right]^{-1}, \quad (42)$$

$$\bar{W}_u(r) = \left\{1 + \frac{\kappa e^{r_c/\sigma}}{4\pi D\sigma}\frac{\sinh K\sigma}{K\sigma}[\chi(1)-\chi(\sigma/r)]\right\}\bar{W}_u(\sigma), \quad (43)$$

with

$$\chi(z) \equiv \int_0^z dy \frac{K\sigma/y}{\sinh(K\sigma/y)}e^{-(r_c/\sigma)y}. \quad (44)$$

It is noted that in the limit of vanishing external field ($K\sigma \to 0$), $\chi(1) \to (\sigma/r_c)(1 - \mathrm{e}^{-r_c/\sigma})$, and $\chi(\sigma/r) \to (\sigma/r_c)(1 - \mathrm{e}^{-r_c/r})$ so that the expression for $\bar{W}_u(r)$ in Eq. (43) reduces to the known exact expression,[10]

$$\lim_{K\sigma\to 0}\bar{W}_u(r) = \left[1 + k_r(e^{r_c/\sigma}-1)\right]^{-1}\left[1 + k_r(e^{r_c/\sigma}\mathrm{e}^{-r_c/r}-1)\right], \quad (45)$$

where $k_r = \kappa/(4\pi D r_c)$. The charge separation probability is enhanced by the external electric field. Up to the linear order in the magnitude of external electric field, the exact solution provides the following expression for the enhancement factor:[2,11]

$$\bar{W}_u(\sigma) \cong \frac{1}{1 + k_r(e^{r_c/\sigma}-1)}\left[1 + \frac{k_r e^{r_c/\sigma}}{1 + k_r(e^{r_c/\sigma}-1)}\frac{K r_c}{2}\right]. \quad (46)$$

By comparison, Eq. (42) for $\bar{W}_u(\sigma)$ gives

$$\bar{W}_u(\sigma) \cong \frac{1}{1+k_r(e^{r_c/\sigma}-1)}\left[1+\frac{k_r e^{r_c/\sigma}}{1+k_r(e^{r_c/\sigma}-1)}\frac{\pi}{\sqrt{6}}\frac{Kr_c}{2}\right]. \tag{47}$$

We thus expect our expression for $\bar{W}_u(\sigma)$ overestimates the charge separation probability at low field strength. However, numerical investigation shows that the above lowest-order expressions are applicable only when the field is very weak, and the overall quality of the solution for extended range of field strength is important.

The expression for $\bar{W}_u(\sigma)$ in Eq. (42) has a similar form as those of previous theories. The corrected Braun's expression[3,4] for $\bar{W}_u(\sigma)$ is given by

$$\bar{W}_u^{BWT}(\sigma) = \left[1+\frac{\kappa(e^{r_c/\sigma}-1)}{4\pi D\sigma}\frac{\sigma}{r_c}\frac{(Kr_c)^{1/2}}{I_1\left(2(Kr_c)^{1/2}\right)}\right]^{-1}, \tag{48}$$

while Seki and Wojcik[5] obtained the following expression:

$$\bar{W}_u^{SW}(\sigma) = \left[1+\frac{\kappa e^{r_c/\sigma}}{4\pi D\sigma}\frac{\sinh K\sigma}{K\sigma}\frac{\sigma}{r_c}\frac{(Kr_c)^{1/2}}{I_1\left(2(Kr_c)^{1/2}\right)}\right]^{-1}. \tag{49}$$

In Eqs. (48) and (49), $I_1(z)$ denotes the modified Bessel function of the first kind.[12] In the limit of vanishing external field, the factor $(Kr_c)^{1/2}/I_1\left(2(Kr_c)^{1/2}\right)$ becomes unity. Hence, it is noted that Eq. (48) also reduces to the known exact expression, but Eq. (49) does only in the large r_c/σ limit. When $r_c/\sigma \gg 1$, both Eqs. (48) and (49) reproduce the linear field dependence of the charge separation probability as given by Eq. (46). It is noteworthy that the first expression for $\bar{W}_u(\sigma)$ [Eq. (25) of Ref. 5] derived by Seki and Wojcik exhibits the correct limiting behavior at low field strength regardless of the magnitude of r_c/σ but behaves unreliable at high field strength.

In Fig. 1, we compare the accuracy of $\bar{W}_u(\sigma)$ expressions in Eqs. (42), (48), and (49) against the exact results calculated numerically by solving Eq. (6) with the finite element method. Actually, for numerical solution of Eq. (6), the presence of δ-function sink term is awkward. Therefore, the δ-function sink term together with the reflecting boundary condition in Eq. (8) was replaced by an equivalent radiative boundary condition

$$\left.\frac{\partial}{\partial r}W_u(r,\mu)\right|_{r=\sigma} = \frac{\kappa(\mu)}{4\pi D\sigma^2}W_u(r,\mu). \tag{50}$$

The key approximations in our theory are those given by Eqs. (19) and (20), which require that $W_u(r,\mu)$ does not depend much on μ. On the physical grounds, these conditions would be satisfied when (i) the anisotropic external electric field is not too large, (ii) the Onsager distance r_c is large compared to the initial charge separation r, and (iii) the inherent reactivity at the contact separation, represented by the parameter κ, is small. For large r_c, the isotropic interaction potential $U_1(r)$ dominates the anisotropic external electric field up to large distance. For small κ, the geminate particles survive long enough that the pair distribution between them deviates little from the isotropic distribution set by the dominant interaction potential $U_1(r)$. Hence the present theory is expected to be more accurate for larger r_c and for smaller K and κ.

Indeed, Fig. 1(a) shows that when $r_c/\sigma = 14$ and $\kappa/4\pi D\sigma = 0.1$, the present theory represented by solid curve as well as the Seki-Wojcik theory represented by dotted-dashed curve produces accurate estimates of the external electric field effect on the charge separation probability. In the figure, the two curves are hardly distinguishable from each other and in good agreement with the exact numerical results represented by filled squares. On the other hand, the dotted curve representing the results of the corrected Braun theory deviates significantly when the strength of the external electric field, measured by $K\sigma$, is large.

However, Fig. 1(b) shows that when $r_c/\sigma = 14$ and $\kappa/4\pi D\sigma = 1.0$, the present theory as well as the Seki-Wojcik theory overestimates the charge separation probability for $K\sigma > 0.5$. Again, the curves drawn for the two theories are hardly distinguishable. The predictions of the corrected Braun theory are still worse. Seki and Wojcik limited the applicability of their theory to the charge separation occurring in low permittivity materials with large r_c. Figure 1(c) shows that when $r_c/\sigma = 5$ and $\kappa/4\pi D\sigma = 0.1$, the Seki-Wojcik theory fails as the value of $K\sigma$ gets larger than 1.5. In fact, for all sets of parameters even with r_c/σ as large as 14, the Seki-Wojcik theory fails at higher $K\sigma$ values. The same trend is observed as in Fig. 1(c); the predicted charge separation probability decreases unphysically at higher $K\sigma$ values. Figure 1(c) shows that the present theory overestimates the charge separation

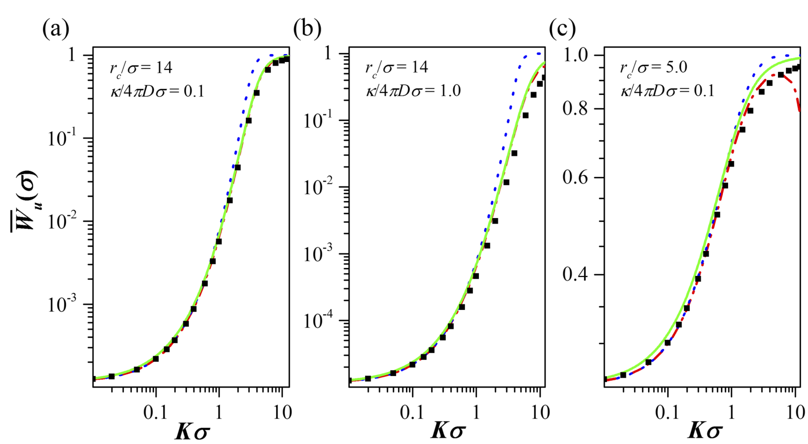

FIG. 1. Charge separation probability $\bar{W}_u(\sigma)$ as a function of $K\sigma$ that measures the strength of external electric field. Exact numerical results are represented by filled squares. Results of the present theory [Eq. (42)], corrected Braun theory [Eq. (48)], and Seki-Wojcik theory [Eq. (49)] are represented by solid, dotted, and dotted-dashed curves, respectively.

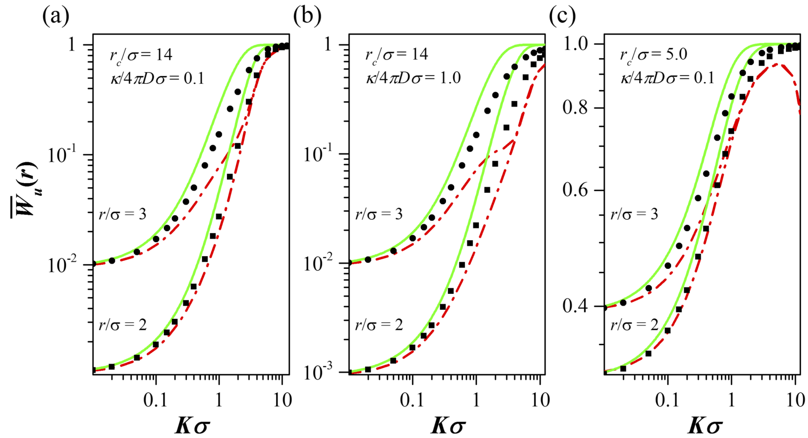

FIG. 2. Charge separation probability $\bar{W}_u(r)$ as a function of $K\sigma$ and initial separation r between geminate particles. Exact numerical results are represented by filled squares for $r/\sigma = 2$ and filled circles for $r/\sigma = 3$. Results of the present theory [Eq. (43)] and Seki-Wojcik theory [Eq. (51)] are represented by solid and dotted-dashed curves, respectively.

probability for the whole range of $K\sigma$. However, for the wider range of $K\sigma$ (i.e., for $K\sigma > 1$), the present theory provides better estimates of charge separation probability than the corrected Braun theory.

In Fig. 2, we displayed the effect of initial separation between geminate particles on the charge separation probability. Seki and Wojcik[5] provided the following expression $\bar{W}_u(r)$:

$$\bar{W}_u^{SW}(r) = 1 - Y(r)[1 - \bar{W}_u^{SW}(\sigma)],$$

$$Y(r) = 1 - \exp\left(-Kr - \frac{r_c}{r}\right)\sum_{n=1}^{\infty}\left(\frac{r}{r_c}\right)^{n-1} \times (Kr_c)^{(n/2)-1} I_n\left(2(Kr_c)^{1/2}\right), \tag{51}$$

where $I_n(z)$ denotes the modified Bessel function of the first kind of integer order n.[12] We compare the accuracy of $\bar{W}_u(r)$ expressions in Eqs. (43) and (51) against the exact results calculated numerically by solving Eq. (6) with the finite element method.

Again, from Fig. 2, we see that the theories provide better estimates of charge separation probability $\bar{W}_u(r)$ when r_c/σ is large and $\kappa/4\pi D\sigma$ is small. However, agreement with the exact numerical results deteriorates as the initial separation r between geminate particles gets larger. The present theory consistently overestimates the charge separation probability in all cases considered, while the Seki-Wojcik theory always underestimates it. For some sets of parameters, the estimates of the Seki-Wojcik theory appear in better agreement with the exact numerical results than those of the present theory. However, the overall trends of variation of $\bar{W}_u(r)$ with respect to r and $K\sigma$ are better predicted by the present theory. Especially, the predictions of the Seki-Wojcik theory become highly unreliable for $r/\sigma \geq 4$. Their estimates for $\bar{W}_u(r)$ collapse to a single value $\bar{W}_u^{SW}(\sigma)$ [Eq. (49)] for large $K\sigma$.

Figure 3 displays the variation of the charge separation probability $\bar{W}_u(\sigma)$ with $K\sigma$ when the interaction between geminate particles is given by the screened Coulomb potential

$$U_1(r) = -\frac{r_c}{r}e^{-\kappa_D r}, \tag{52}$$

where κ_D^{-1}, called the Debye-Hückel length, measures the thickness of the ionic atmosphere around a charged particle. In this case, from Eqs. (24) and (25), the separation probability of a geminate charge pair with initial separation r is given by

$$\bar{W}_u(\sigma) = \left\{1 + \frac{\kappa \exp[(r_c/\sigma)e^{-\kappa_D\sigma}]}{4\pi D\sigma}\frac{\sinh K\sigma}{K\sigma}\chi(1)\right\}^{-1}, \tag{53}$$

$$\bar{W}_u(r) = \left\{1 + \frac{\kappa \exp[(r_c/\sigma)e^{-\kappa_D\sigma}]}{4\pi D\sigma}\frac{\sinh K\sigma}{K\sigma}[\chi(1) - \chi(\sigma/r)]\right\} \times \bar{W}_u(\sigma), \tag{54}$$

with $\chi(z)$ given by

$$\chi(z) \equiv \int_0^z dy\frac{K\sigma/y}{\sinh(K\sigma/y)}\exp[-(r_c/\sigma)ye^{-\kappa_D\sigma/y}]. \tag{55}$$

Again, in the limit of vanishing external field ($K\sigma \to 0$), the expression for $\bar{W}_u(r)$ in Eq. (54) reduces to the exact

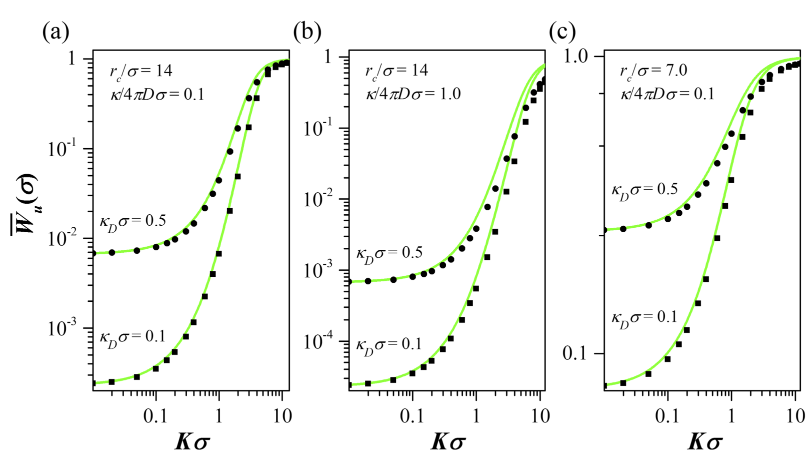

FIG. 3. Charge separation probability $\bar{W}_u(\sigma)$ as a function of $K\sigma$ when the interaction between geminate particles is given by the screened Coulomb potential in Eq. (52). Exact numerical results are represented by filled squares for $\kappa_D\sigma = 0.1$ and filled circles for $\kappa_D\sigma = 0.5$, and results of the present theory [Eq. (53)] are represented by solid curves.

expression.[10] As the Debye-Hückel length κ_D^{-1} decreases, the attractive interaction between geminate particles diminishes rapidly with increasing separation so that the charge separation probability increases with κ_D.

Figure 3 shows that the present theory provides reliable estimates of $\bar{W}_u(\sigma)$ over the whole range of $K\sigma$ when the Onsager distance r_c is large and the inherent reactivity at the contact separation is small; in Fig. 3(a), $r_c/\sigma = 14$ and $\kappa/4\pi D\sigma = 0.1$. However, as in the case with bare Coulomb interaction (see Fig. 1), the accuracy deteriorates progressively with increasing κ [Fig. 3(b)] and decreasing r_c [Fig. 3(c)]. Also when the Debye-Hückel length κ_D^{-1} is small and the initial separation between geminate particles is large, the influence of isotopic interaction between the particles diminishes and the present estimates of charge separation probability may become less accurate.

B. Systems involving a long-range reaction sink

When the recombination of geminate particles occurs via electron transfer, one may use the Marcus theory of electron transfer reactions to model the reaction sink functions. The rate for the nonadiabatic electron transfer rate is given by[13]

$$S_R(r,\mu) = \frac{2\pi}{\hbar}\frac{|V|^2}{(4\pi\lambda k_BT)^{1/2}}\exp\left(-\frac{(\Delta G+\lambda)^2}{4\lambda k_BT}\right). \tag{56}$$

Here V and ΔG are the matrix element of the interaction Hamiltonian and the free energy difference between the donor state and the acceptor state, respectively, and λ is the reorganization energy. V depends very sharply on the distance between the donor and acceptor. One usually assumes the following functional form for the distance dependence of V:

$$|V|^2 = |V_0|^2 e^{-\alpha(r-\sigma)}, \tag{57}$$

with V_0 denoting the coupling at the contact distance. As will be described below, ΔG is a function of r and μ in general. For simplicity, however, we will first take it to be constant and consider a reaction sink function expressed as

$$S_R(r) = \nu_0\,\bar{S}_R(y = r/\sigma) \;\text{ with }\; \bar{S}_R(y) = e^{-A^2/(4\bar{\lambda})}e^{-\alpha\sigma(y-1)}, \tag{58}$$

where $\nu_0 = (2\pi/\hbar)|V_0|^2/(4\pi\lambda k_BT)^{1/2}$, $\bar{\lambda} = \lambda/k_BT$, and $A = \Delta G_0/(k_BT) + \bar{\lambda}$. A typical value of the prefactor ν_0 in the case of organic photovoltaics is about $10^{13}\ \mathrm{s}^{-1}$.[14] With the interaction potential $U_1(r) = -r_c/r$, we then obtain from Eq. (39) in Sec. II the following expression for the charge separation probability:

$$\bar{W}_u(x) = \left[1 + \frac{\nu_0\sigma^2}{D}\frac{1}{\bar{W}_u^\delta(x)}\left(\chi(x^{-1})\int_1^x dy\frac{y^2\bar{\Lambda}(y)}{e^{U_1(\sigma y)}}\bar{W}_u^\delta(y) + \int_x^\infty dy\frac{y^2\bar{\Lambda}(y)}{e^{U_1(\sigma y)}}\chi(y^{-1})\bar{W}_u^\delta(y)\right)\right]^{-1}, \tag{59}$$

where $x = r/\sigma$ and $\chi(z)$ is the function defined in Eq. (44). $\bar{\Lambda}(y)$ and $\bar{W}_u^\delta(y)$ are given by

$$\bar{\Lambda}(y) = \bar{S}_R(y)\frac{\sinh K\sigma y}{K\sigma y}, \tag{60}$$

$$\bar{W}_u^\delta(y) = 1 + \frac{\Lambda_{rx}}{4\pi D\sigma}e^{-U_1(\sigma)}[\chi(1) - \chi(y^{-1})], \tag{61}$$

with

$$\frac{\Lambda_{rx}}{4\pi D\sigma} = \frac{\sigma^2\nu_0}{D}e^{-r_c/\sigma}\frac{\sinh K\sigma}{K\sigma}\int_1^\infty dy\, y^2 e^{(r_c/\sigma)/y}\bar{S}_R(y). \tag{62}$$

For $x = 1$ (i.e., $r = \sigma$), Eq. (59) reduces to

$$\bar{W}_u(1) = \left[1 + \frac{\nu_0\sigma^2}{D}\frac{1}{\bar{W}_u^\delta(1)}\int_1^\infty dy\frac{y^2\bar{\Lambda}(y)}{e^{U_1(\sigma y)}}\chi(y^{-1})\bar{W}_u^\delta(y)\right]^{-1}. \tag{63}$$

In Fig. 4, we compare the accuracy of $\bar{W}_u(r)$ expressions given by Eqs. (59)–(63) against the exact results calculated numerically by solving Eq. (6) with the finite element method. To calculate $\bar{W}_u(r)$, we need seven dimensionless parameters, $K\sigma$, r_c/σ, $\nu_0\sigma^2/D$, r/σ, A, $\bar{\lambda}$, and $\alpha\sigma$. Figure 4 displays the influence of the four parameters $K\sigma$, r_c/σ, $\nu_0\sigma^2/D$, and r/σ on the accuracy of the approximate analytic expression. We fixed the values of the other three parameters as $A = 20$, $\bar{\lambda} = 10$, and $\alpha\sigma = 2$, which are the typical values in the case of organic photovoltaics.

Again, when the Onsager distance r_c is large and the inherent reactivity parameter $\nu_0\sigma^2/D$ is small, the agreement of analytic estimates of $\bar{W}_u(\sigma)$ with the exact numerical results is excellent. This exemplifies the utility of our solution method for Fredholm integral equations of the second kind.[6,7] However, as in the δ-function sink case, the agreement deteriorates as the Onsager distance r_c decreases [Fig. 4(c)] and the initial separation r between geminate particles increases. This is

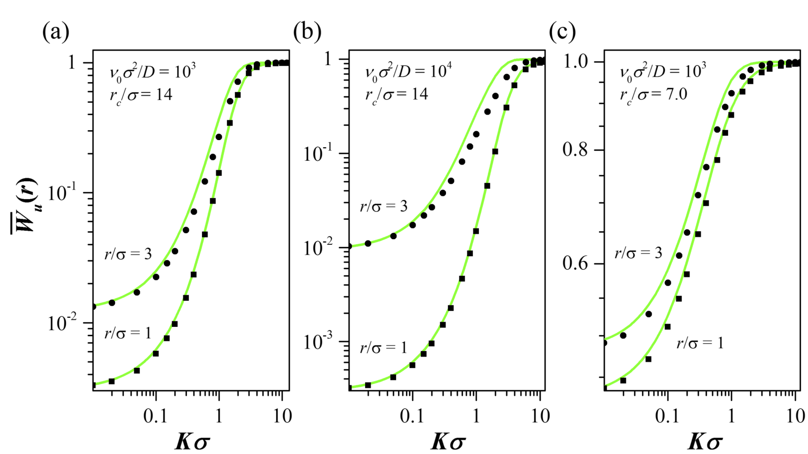

FIG. 4. Charge separation probability $\bar{W}_u(r)$ as a function of $K\sigma$ and initial separation r between geminate particles when the reaction sink function is given by Eq. (58). Exact numerical results are represented by filled squares for $r/\sigma = 1$ and filled circles for $r/\sigma = 3$. Results of the present theory [Eq. (59)] are represented by solid curves.

because the key approximations given by Eqs. (19) and (20) become less accurate as the effects of anisotropic external field dominate the isotropic central interaction potential.

As mentioned above, ΔG in Eq. (56) depends on r and μ in general. Wang and Suna[14] proposed a simple model for the field dependence of ΔG as expressed by

$$\frac{\Delta G(r,\mu)}{k_BT} = \frac{\Delta G_0}{k_BT} + \frac{r_c}{r} + Kr\mu, \tag{64}$$

where ΔG_0 denotes an intrinsic free energy difference between the donor state and the acceptor state. With Eq. (64), the reaction sink function can be represented as

$$S_R(r,\mu) = \nu_0\,\bar{S}_R(y = r/\sigma, \mu),$$

with

$$\bar{S}_R(y,\mu) = \exp\left(-\frac{1}{4\bar{\lambda}}\left[A + \frac{r_c/\sigma}{y} + K\sigma y\mu\right]^2\right) e^{-\alpha\sigma(y-1)}, \tag{65}$$

where ν_0, $\bar{\lambda}$, and A are the same quantities as defined after Eq. (58). However, this simple model poses a problem when either r_c/σ or $K\sigma$ is large; the charge recombination is predicted to occur faster at a separation larger than the contact distance σ, which seems rather unphysical. As shown in Fig. 5, when $r_c/\sigma = 14$ and $K\sigma \geq 1.5$, the charge separation probability $\bar{W}_u(r)$ calculated numerically with Eq. (65) turns out to be a non-monotonic function of the initial separation between geminate particles; values of other parameters used for the numerical solution are $\nu_0\sigma^2/D = 10^3$, $A = 10$, $\bar{\lambda} = 10$, and $\alpha\sigma = 2$. It is obvious that one needs to consider more carefully the distance-dependence of ΔG, as well as the interaction Hamiltonian V and the reorganization energy λ, in the presence of strong external electric fields. However, the primary purpose of the present work is to formulate an analytic theory of diffusion-influenced charge separation for a given model of reaction sink function, and we will simply employ the model function given by Eq. (65) to test the accuracy of the approximate expression for $\bar{W}_u(r)$ derived in Sec. II.

With the interaction potential $U_1(r) = -r_c/r$ and the reaction sink function in Eq. (65), we obtain from Eq. (39) the same expression in Eq. (59) for the charge separation probability. $\bar{W}_u^{\delta}(y)$ appearing in Eq. (59) is also given by the same expression as in Eq. (61), but now $\bar{\Lambda}(y)$ and $\Lambda_{rx}/(4\pi D\sigma)$ are given by

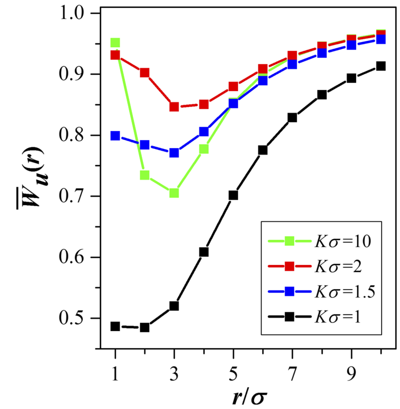

FIG. 5. Dependence of charge separation probability $\bar{W}_u(r)$ on the initial separation r/σ between geminate particles. The results are the exact ones calculated numerically from Eq. (6) with the model sink function in Eq. (65). Values of parameters used for the numerical solution are $r_c/\sigma = 14$, $\nu_0\sigma^2/D = 10^3$, $A = 10$, $\bar{\lambda} = 10$, and $\alpha\sigma = 2$.

$$\begin{aligned}\bar{\Lambda}(y) = {}& \frac{1}{2}\sqrt{\pi\bar{\lambda}}\frac{e^{\bar{\lambda}-B(y)-\alpha\sigma(y-1)}}{K\sigma y}\left\{\mathrm{erfc}\left[\frac{B(y)-K\sigma y-2\bar{\lambda}}{2\sqrt{\bar{\lambda}}}\right]\right.\\ &\left. -\,\mathrm{erfc}\left[\frac{B(y)+K\sigma y-2\bar{\lambda}}{2\sqrt{\bar{\lambda}}}\right]\right\},\end{aligned} \tag{66}$$

$$\begin{aligned}\frac{\Lambda_{rx}}{4\pi D\sigma} = {}& \frac{1}{2}\frac{\sigma^2\nu_0}{D}\frac{\sqrt{\pi\bar{\lambda}}}{K\sigma}e^{-r_c/\sigma}\int_1^{\infty} dy\, y\\ &\times \exp\left(\frac{(r_c/\sigma)}{y} - \alpha\sigma(y-1) + \frac{\bar{\lambda}}{y^2} - \frac{B(y)}{y}\right)\\ &\times\left\{\mathrm{erfc}\left[\frac{B(y)-K\sigma y-2\bar{\lambda}/y}{2\sqrt{\bar{\lambda}}}\right]\right.\\ &\left. -\,\mathrm{erfc}\left[\frac{B(y)+K\sigma y-2\bar{\lambda}/y}{2\sqrt{\bar{\lambda}}}\right]\right\},\end{aligned} \tag{67}$$

where erfc(z) is the complementary error function[12] and $B(y) \equiv A + (r_c/\sigma)y^{-1}$.

In Fig. 6, we compare the accuracy of $\bar{W}_u(r)$ expressions given by Eqs. (59), (61), (66), and (67) against the exact results calculated numerically by solving Eq. (6) with the finite element method. To calculate $\bar{W}_u(r)$, we need seven dimensionless parameters, $K\sigma$, r_c/σ, $\nu_0\sigma^2/D$, r/σ, A, $\bar{\lambda}$, and $\alpha\sigma$. Figure 6 displays the influence of the four parameters $K\sigma$,

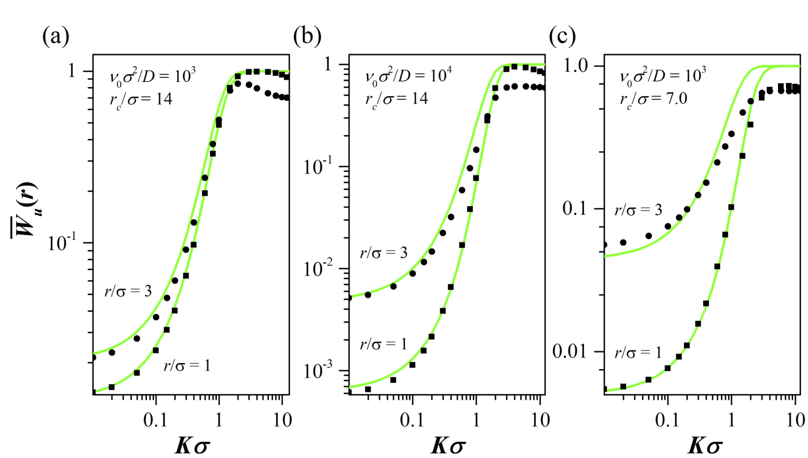

FIG. 6. Charge separation probability $\bar{W}_u(r)$ as a function of $K\sigma$ and initial separation r between geminate particles when the reaction sink function is given by Eq. (65). Exact numerical results are represented by filled squares for $r/\sigma = 1$ and filled circles for $r/\sigma = 3$. Results of the present theory [Eqs. (59), (61), (66), and (67)] are represented by solid curves.

r_c/σ, $\nu_0\sigma^2/D$, and r/σ on the accuracy of the approximate analytic expression. We fixed the values of the other three parameters as $A = 10$, $\bar{\lambda} = 10$, and $\alpha\sigma = 2$.

We note that $\bar{W}_u(r)$ obtained by numerical solution of Eq. (6) behaves unphysically for large $K\sigma$. It reaches a plateau value less than unity, and even decreases when $K\sigma$ is further increased. As mentioned above, this indicates the deficiency of the simple reaction sink model given by Eq. (65). Apart from the region of large $K\sigma$, the agreement of analytic estimates of $\bar{W}_u(r)$ with the exact numerical results are excellent, especially when the Onsager distance r_c is large and the inherent reactivity parameter $\nu_0\sigma^2/D$ is small. Again, the agreement deteriorates as the Onsager distance r_c decreases [Fig. 6(c)] and the initial separation r between geminate particles increases.

IV. CONCLUDING REMARKS

We presented a closed analytic expression for the separation probability of geminate charged particles under the influence of constant external electric field. Compared to the well-known Braun's expression,[3] as corrected by Wojcik and Tachiya,[4] and the recently proposed Seki and Wojcik's expression, our expression provides overall more reliable estimates for the charge separation probability. In addition, we have extended the applicability of expression to the cases in which the recombination of geminate particles needs to be described by an anisotropic, long-range reaction sink function and the interaction between the geminate particles is given by arbitrary central potential energy.

The recombination of geminate charge pairs generated by radiation has been investigated intensively, especially in an effort to design more efficient solar cells.[2–5,15,16] The spherical geminate particles considered in the present theory can be a simple model of a charge-transfer exciton created in an organic photovoltaics. For application to the more realistic model system, however, one needs to consider the charge separation process occurring in heterogeneous media.[11] We believe that the mathematical approach presented in this work would help develop an analytical solution to such complex problems.

ACKNOWLEDGMENTS

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[1] L. Onsager, Phys. Rev. **54**, 554 (1938).
[2] J. Noolandi and K. M. Hong, J. Chem. Phys. **70**, 3230 (1979).
[3] C. L. Braun, J. Chem. Phys. **80**, 4157 (1984).
[4] M. Wojcik and M. Tachiya, J. Chem. Phys. **130**, 104107 (2009).
[5] K. Seki and M. Wojcik, J. Phys. Chem. C **121**, 3632 (2017).
[6] S. Lee, C. Y. Son, J. Sung, and S. Chong, J. Chem. Phys. **134**, 121102 (2011).
[7] C. Y. Son, J. Kim, J.-H. Kim, J. S. Kim, and S. Lee, J. Chem. Phys. **138**, 164123 (2013).
[8] C. Gardiner, *Stochastic Methods*, 4th ed. (Springer, Berlin, 2009).
[9] K. M. Hong and J. Noolandi, J. Chem. Phys. **69**, 5026 (1978).
[10] M. Tachiya, J. Chem. Phys. **69**, 2375 (1978).
[11] M. Wojcik, A. Nowak, and K. Seki, J. Chem. Phys. **146**, 054101 (2017).
[12] M. Abramowitz and I. A. Stegun, *Handbook of Mathematical Functions* (Dover, New York, 1972).
[13] A. Nitzan, *Chemical Dynamics in Condensed Phases* (Oxford, UK, 2006).
[14] Y. Wang and A. Suna, J. Phys. Chem. B **101**, 5627 (1997).
[15] T. M. Clarke and J. R. Durrant, Chem. Rev. **110**, 6736 (2010).
[16] K. M. Pelzer and S. B. Darling, Mol. Syst. Des. Eng. **1**, 10 (2016).